The Journal of Open Source Software

Mashpit: sketching out genomic epidemiology

Tongzhou Xu [1], **Henk C. den Bakker** [1], **Xiangyu Deng** [1], and **Lee S. Katz** [1,2]

1 Center for Food Safety, University of Georgia, Griffin, GA, United States of America **2** Enteric Diseases Laboratory Branch (EDLB), Centers for Disease Control and Prevention, Atlanta, GA, United States of America

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Charlotte Soneson

Reviewers:

- @hkaspersen
- @mberacochea

Submitted: 09 September 2024
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

We are in the era of genomic epidemiology. The surveillance of many transmissible diseases is increasingly being conducted through whole genome sequencing of pathogenic agents. One notable example is *Salmonella*, a major foodborne pathogen routinely sequenced by surveillance programs such as PulseNet. Large volumes of *Salmonella* genomes from these programs are deposited in database systems including NCBI (Nadon et al., 2017). These publicly available genomes can be analyzed in a variety of ways such as serotyping (Zhang et al., 2019), multilocus sequence typing (MLST) (Zhou et al., 2020), and single nucleotide polymorphism (SNP) typing (Katz et al., 2017). These analyses provide important laboratory evidence for outbreak surveillance and investigation.

As of August 2024, there are more than 600 thousand *Salmonella* genomes and more than half a million other pathogen genomes at NCBI Pathogen Detection (https://www.ncbi.nlm.nih.gov/pathogens). These numbers are expected to increase dramatically and therefore faster methods are needed.

There have been some major advances to scale up bioinformatic analyses to large volumes of pathogenic genomes. One approach is to provide centralized resources that integrate data and analytical tools. For example, Pathogen Detection combines information from three databases: SRA, GenBank, and BioSample. About once a day, it compares all genomes of a given taxon, separates all genomes into individual clusters using MLST, and then creates a phylogeny for each cluster using SNP analysis. This method is quite comprehensive, but it relies on each sample being public, and it cannot be executed locally.

Another approach is to provide new tools for decentralized and customized manipulation of genomics resources. We observed that an algorithm for genomics called Min-Hash is well positioned for this purpose. A commonly used software for Min-Hash is called Mash (Ondov et al., 2016). Querying with Mash can be about 3 orders of magnitude faster than other common methods like Basic Local Alignment Search Tool (BLAST) and can have a smaller disk footprint (Camacho et al., 2009). Therefore it can be run on more common scientific workstations.

We present Mashpit, a new rapid genomic epidemiology platform to query against these large groups of genomes on a local computer.

Statement of need

Querying a sample against these magnitudes of genomes is becoming less sustainable, especially for smaller laboratories. Currently, GISAID and NCBI are staying ahead of the curve by producing a global tree of each organism every day. This requires herculean efforts, cutting

edge algorithms, and powerful computers. However, smaller laboratories usually have a scientific workstation or similar equipment, much different than a cluster computing system.

We note that for some organisms like Salmonella, queries can be of a sensitive nature. For example, harboring isolates in food production environments that are related to outbreak isolates is often perceived as a potential liability by food establishments, therefore thwarting the efforts to use and share the genomes of these organisms.

To address any needs for speed and sensitivity, we created Mashpit. Mashpit queries genomes locally using Mash, thereby achieving speedy results while keeping any sensitive queries offline.

Mashpit design

Mashpit is comprised of three major parts: A min-hash database, its associated metadata, and the min-hash querying.

The database is created with an interface to Mash, called Sourmash (Brown & Irber, 2016). Each genome is imported by sketching it and adding it to a Sourmash signature database. Each genome can also have an entry in the associated metadata. These data include date of isolation, geography, host age range, and other information that could be useful in an epidemiological investigation. Mashpit can build a species database from NCBI Pathogen Detection, termed a Mashpit taxon database or a custom database from user-provided genomes. The Mashpit taxon database is based on the available pathogen species on Pathogen Detection. For each SNP cluster of one species on Pathogen Detection, the set of all genomes in an SNP cluster is defined as:

$$G = \{g_1, g_2, ..., g_n\}$$

where n is the number of genomes in the cluster. The centroid genome g_c is calculated as:

$$g_c = \underset{g_i \in G}{argmin} \sum_{j=1}^{n} d(g_i, g_j)$$

Where $d(g_i, g_j)$ is the distance between two genomes. By default, Mashpit will download the latest SNP cluster for specified species and uses a kmer size of 31 and kmer number of 1000 for sketching the genomes.

To evaluate the performance of Mashpit, we tested Mashpit on a server that runs Ubuntu 20.04.2 with an Intel Xeon CPU E5-2697 v4 2.30GHz and 256GB RAM. The elapsed time of building a database and running a query was calculated for four of the major foodborne pathogens: *Salmonella*, *Listeria*, *E. coli*, and *Campylobactor*.

With the database and its metadata complete, a user could perform a query. The query is an assembly fasta file, which is then sketched and compared against the signature database. The query then returns a tab delimited spreadsheet, sorted by Mash distance and a phylogenetic tree based on the Mash distance. All associated metadata are included in the spreadsheet.

Performance

To evaluated the performance of Mashpit, we constructed Mashpit taxon databases for *Salmonella*, *Listeria*, *E. coli*, and *Campylobacter* using NCBI pathogen detection SNP clusters versioned before January 2024. We randomly selected 1000 newly added genomes for each species added to NCBI pathogen detection after January 2024. Subsequently, we queried these genomes against Mahpit taxon databases and recorded the time taken for each step (Figure 1). We also compared the query results with the true SNP cluster of the query genomes. We calclated the proportion of true SNP clusters appearing among the top hits at various thresholds (Figure 1). The 'threshold' indicates whether the correct SNP cluster is among the top 'threshold number' of query hits. For instance, a threshold of 25 indicates that the correct

cluster is among the top 25 hits. Our findings indicate that *Salmonella* demonstrated a 70% success rate in having the true cluster within the top 25 hits while *Campylobacter* showed a success rate of approximately 90%.

Discussion

We present Mashpit, a rapid genomic epidemiology platform. Due to the underlying algorithm Min-Hash, it is exceedingly fast. It also has such a small hard drive and computational footprint that it can basically be used on common scientific workstations. However, we note that the Mash distance does not correlate well to well-established distances such as MLST. Therefore we recommend that this platform is used as a first-pass to filter unrelated samples before using a more established protocol such as MLST. In conclusion, we believe that Mashpit is an essential genomic epidemiology tool.

Figures

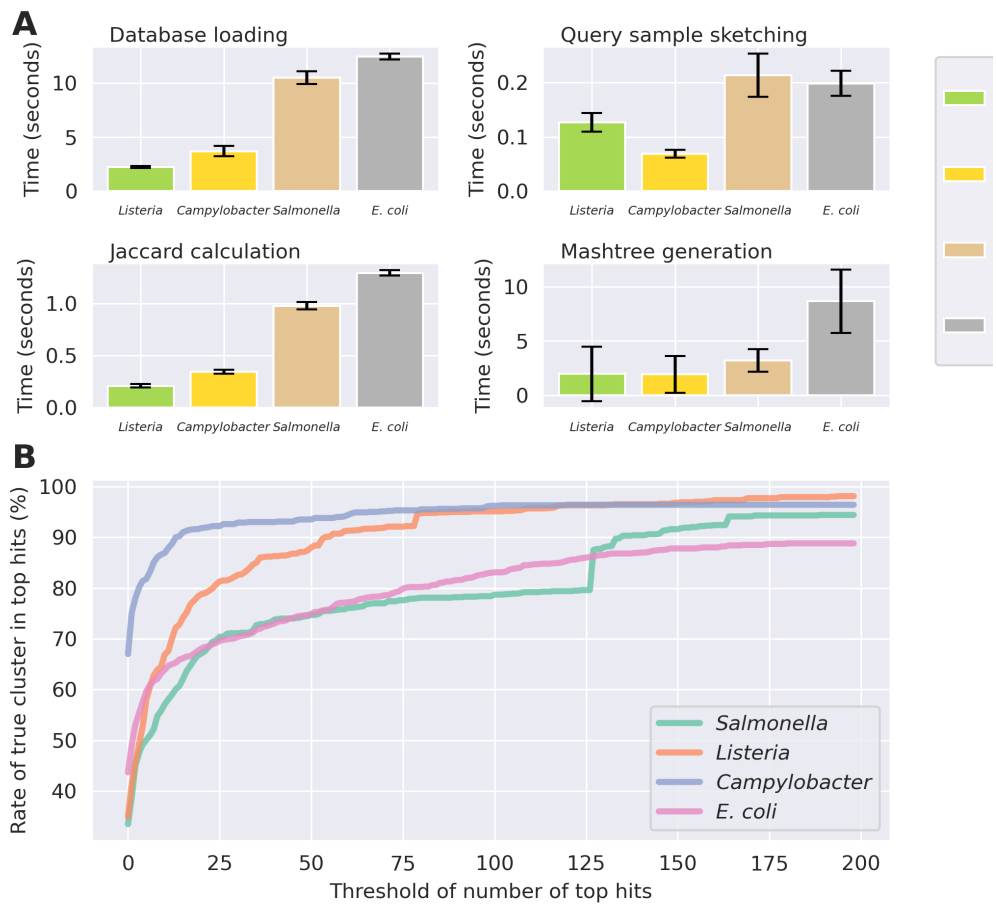

Figure 1: Performance of Mashpit query. A. Average time of a query against four Mashpit taxon databases. B. Rate of true SNP clusters in top rate with different thresholds.

Acknowledgements

Financial support for the development of Mashpit was provided by the Center for Food Safety at the University of Georgia, United States of America. The findings and conclusions in this report are those of the authors and do not necessarily represent the official position of the Centers for Disease Control and Prevention.

References

Brown, C. T., & Irber, L. (2016). Sourmash: A library for MinHash sketching of DNA. *Journal of Open Source Software*, *1*(5), 27. https://doi.org/10.21105/joss.00027

Camacho, C., Coulouris, G., Avagyan, V., Ma, N., Papadopoulos, J., Bealer, K., & Madden, T. L. (2009). BLAST+: Architecture and applications. *BMC Bioinformatics*, *10*(1), 1–9. https://doi.org/10.1186/1471-2105-10-421

Katz, L. S., Griswold, T., Williams-Newkirk, A. J., Wagner, D., Petkau, A., Sieffert, C., Van Domselaar, G., Deng, X., & Carleton, H. A. (2017). A comparative analysis of the lyve-SET phylogenomics pipeline for genomic epidemiology of foodborne pathogens. *Frontiers in Microbiology*, *8*, 375. https://doi.org/10.3389/fmicb.2017.00375

Nadon, C., Van Walle, I., Gerner-Smidt, P., Campos, J., Chinen, I., Concepcion-Acevedo, J., Gilpin, B., Smith, A. M., Kam, K. M., Perez, E., & others. (2017). PulseNet international: Vision for the implementation of whole genome sequencing (WGS) for global food-borne disease surveillance. *Eurosurveillance*, *22*(23), 30544. https://doi.org/10.2807/1560-7917.es.2017.22.23.30544

Ondov, B. D., Treangen, T. J., Melsted, P., Mallonee, A. B., Bergman, N. H., Koren, S., & Phillippy, A. M. (2016). Mash: Fast genome and metagenome distance estimation using MinHash. *Genome Biology*, *17*(1), 1–14. https://doi.org/10.1186/s13059-016-0997-x

Zhang, S., Bakker, H. C. den, Li, S., Chen, J., Dinsmore, B. A., Lane, C., Lauer, A., Fields, P. I., & Deng, X. (2019). SeqSero2: Rapid and improved salmonella serotype determination using whole-genome sequencing data. *Applied and Environmental Microbiology*, *85*(23), e01746–19. https://doi.org/10.1128/aem.01746-19

Zhou, Z., Alikhan, N.-F., Mohamed, K., Fan, Y., Achtman, M., Brown, D., Chattaway, M., Dallman, T., Delahay, R., Kornschober, C., & others. (2020). The EnteroBase user's guide, with case studies on salmonella transmissions, yersinia pestis phylogeny, and escherichia core genomic diversity. *Genome Research*, *30*(1), 138–152. https://doi.org/10.1101/gr.251678.119

DRAFT